

MADHAV CHERUPILIL SAJEEV

Email: madhavcs123@gmail.com

RESEARCH INTERESTS

My interests lie in Applied and Computational Topology/Geometry and its applications.

EDUCATION

Montana State University
PhD in Computer Science

Bozeman, Montana, USA
Aug 2026 - Present

Ecole Polytechnique, IP Paris
MPRI Parisian Master of Research in Computer Science

Paris, France
Completed 2026 September

Chennai Mathematical Institute
BSc (Hons) Mathematics and Computer Science

Chennai, Tamil Nadu, India
Completed 2024 April

PREPRINTS

- Classification of Temporal Graphs using Persistent Homology (Joint work with Samrik Chowdhury, Siddharth Pritam and Rohit Roy) [arXiv:502.10076](https://arxiv.org/abs/502.10076)

INTERSHIPS

- Computational Conley Index Theory (Under the supervision of Alxendra Bac and Aldo Gonzalez-Lorenzo)
- Triangulating Submanifolds with Boundaries (Under the supervision of Mathijs Wintraecken)

RELEVANT COURSE WORK

Mathematics: Algebraic Topology of Manifolds, Characteristic Classes, Homotopy Theory, Homology Cohomology and Sheafs, Tropical Aspects of Enumerative Geometry, Riemann Surfaces, Algebraic Topology, Differential Equations, Topology, Abstract Algebra, Real and Complex Analysis.

Computer Science: Topological Data Analysis, Algorithmic Geometry, Homotopy Type Theory. Domains Categories and Games, Symbolic Dynamics, Well Quasi Orders, Control Systems, Interactive Proofs Undecidability in Algebra and Topology, Foundations of Machine Learning.

TALKS

- Presentable Categories (∞ -Category Theory Seminar at Sorbonne)
- Path Contractibility Problem (Student Seminar at CMI)
- Applications of TDA (TDA Workshop at IIT Madras)
- Temporal Graph Filtration (Applied Topology and Complex Nextwork Workshop at CMI)
- Subgraph Isomorphism in Temporal Networks (Applied Topology Learning Seminar)

CONFERENCES AND WORKSHOPS ATTENDED

- Higher Order Interactions at the Crossroads of geometry and Topology (Workshop at Paris Saclay)
- SGI 2024 Tutorial Week (Online Workshop at MIT)
- Applied Topology and Complex Nextwork (Workshop at CMI)
- Applied Topology (Seminar at CMI)

SKILLS

Technical Skills Python, Latex